Neural Networks for Sequential Learning: Recurrent Neural Network (RNN)

1

So Far … Static Pattern Learning

- Feed forward neural networks
  - Fully connected neural network (FCNN)
  - Convolutional neural network (CNN)
- The inputs to the CNN and FCNN are always of a fixed size

FCNN → d-dimn. vector.

LeNet → 24x24 / 36x36 VGG → 224x224

- Each input is independent to the previous or future input
  - Example: Image classification:

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Towards Sequential Learning

- Feed forward neural networks
  - Fully connected neural network (FCNN)
  - Convolutional neural network (CNN)
- The inputs to the CNN and FCNN are always of a fixed size
- Each input is independent to the previous or future input

- In many applications the input is not of a fixed size
- Further successive inputs may not be independent of each other

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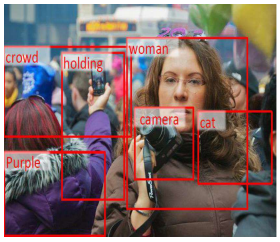
Image Classification

[1]



[2]

- The inputs to the CNN and fully connected neural network (FCNN) are always of a fixed size
- Each input is independent to the previous or future input
- In many applications the input is not of a fixed size
- Further successive inputs may not be independent of each other

[1] O. Vinyals, A. Toshev, S. Bengio and D. Erhan, "Show and tell: Lessons learned from the 2015 MSCOCO Image Captioning Challenge," IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 39, no.4, pp.652-663, April 2017.

[2] Fang et al., "From captions to visual concepts and back, CVPR, 2015.

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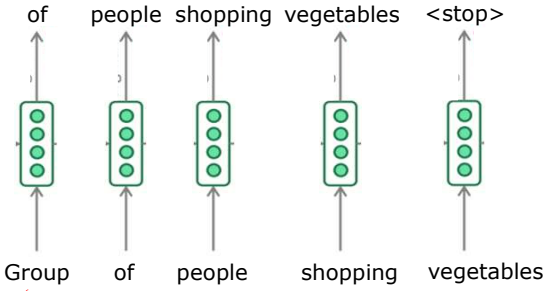
Language Modeling

- Consider the problem of language modelling: Natural sentence generation
  - Given $t-i$ words predict the t^{th} word
- Example: Generate a sentence – "Group of people shopping vegetables"
  - A word *shopping* is predicted given the words *Group*, *of*, *people*



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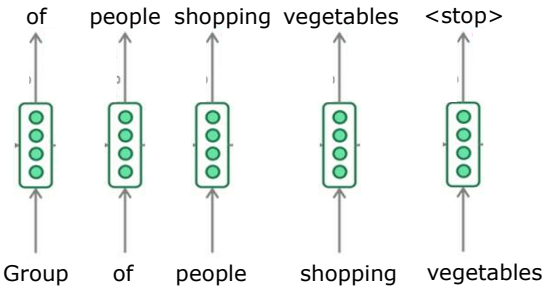
Sequence Learning Problem: Language Modeling

- Natural sentence generation
  - Given $t-i$ words predict the t^{th} word
- Each input corresponding to one time step



- Inputs are dependent on each other
- Lengths of the inputs are not fixed (sentences could have arbitrary number of words)
- Each network is performing same task (input: word, output: word)
- **Sequence learning problem**:
  - Look at the sequence of (dependent) inputs and produce output(s)

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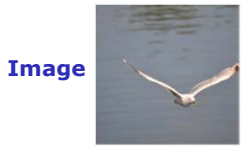
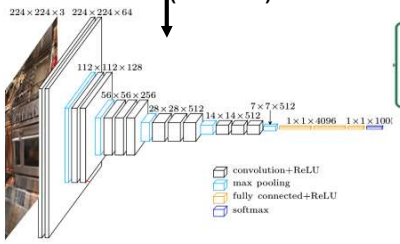
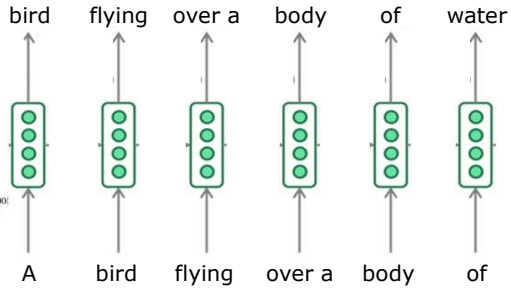
Sequence Learning Problem: Natural Image Caption Generation

- Generate a sentence given an image



A bird flying over a body of water [3]





Natural language generation

[3] Kelvin Xu, Jimmy Ba, Ryan Kiros, Kyunghyun Cho, Aaron Courville, Ruslan Salakhudinov, Rich Zemel, Yoshua Bengio, "Show, Attend and Tell: Neural Image Caption Generation with Visual Attention", in *Proceedings of the 32nd International Conference on Machine Learning*, PMLR vol. 37, pp. 2048-2057, 2015.

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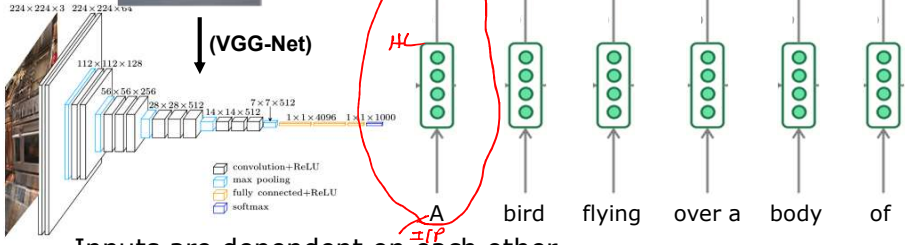
Sequence Learning Problem: Natural Image Caption Generation

A bird flying over a body of water [3]

Sequence learning problem



- Inputs are dependent on each other
- Lengths of the inputs are not fixed (sentences could have arbitrary number of words)
- Each network is performing same task (input: word, output: word)

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Sequence Learning Problem

- Sometimes we may not be interested in producing an output at every stage
- Instead we would look at the full sequence and then produce an output
- For example, consider the task of video classification or activity recognition from the video
- Look at the entire sequence and predict the activity being performed

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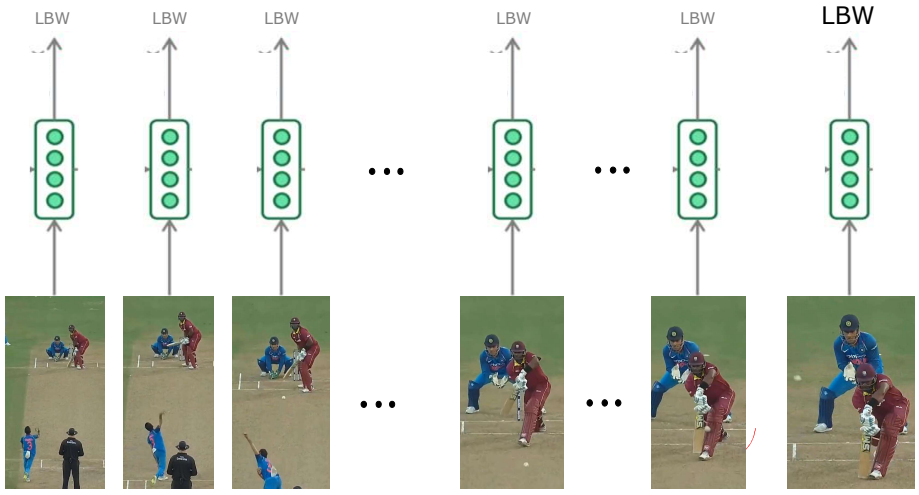
Sequence Learning Problem: Video Classification or Activity Recognition



- The prediction clearly does not depend only on the last frame but also on some frames which appear before
- Network is performing the same task at each step (input: frame (Image), output: Activity class)-Don't care about intermediate outputs

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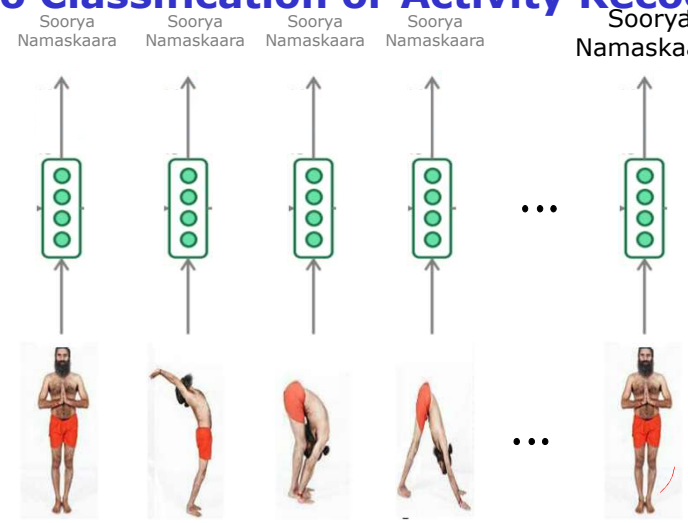
Sequence Learning Problem: Video Classification or Activity Recognition



- The prediction clearly does not depend only on the last frame but also on some frames which appear before
- Network is performing the same task at each step (input: frame (Image), output: Activity class)-Don't care about intermediate outputs

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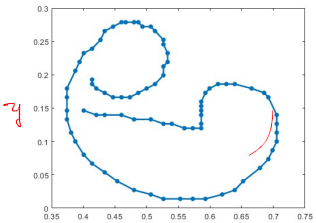
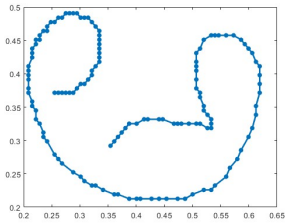
Sequence Learning Problem: Handwritten Character Recognition

- A character is represented as a sequence of 2-dimentional points in one stroke (from pen-down to pen-up)

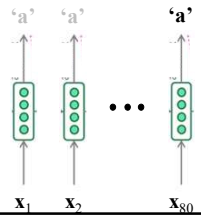


Sequence of 80 points $(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_{80})$



Sequence of 138 points $(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_{138})$

Character 'a' from Kannada/Telugu



- The prediction clearly does not depend only on the last feature vector in the sequence but also on some feature vectors which appear before
- Network is performing the same task at each step (input: 2d-data point, output: character class)-Don't care about intermediate outputs

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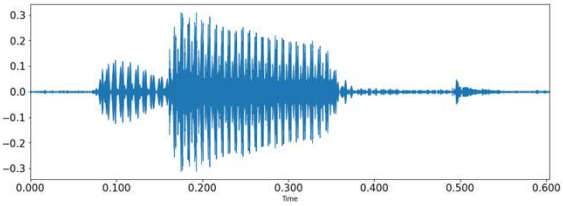
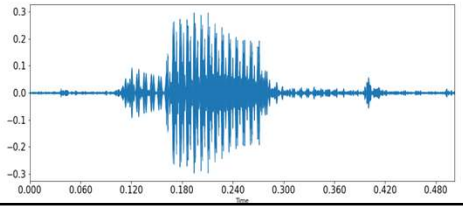
Sequence Learning Problem: Spoken Word Recognition

- Context: Limited vocabulary word recognition
  - Example: Systems working on the voice commands
- Each spoken word is a sequence of sounds
  - Example spoken word: "deep"
  - Sequence of sounds: /d/ /e/ /p/
- Duration of the utterances of the same word varies from each other





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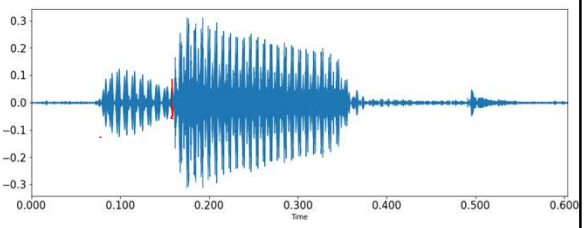
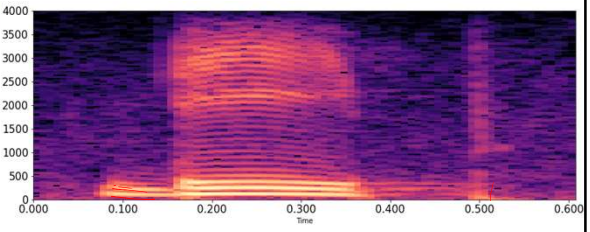
Sequence Learning Problem: Spoken Word Recognition

- Representing the speech in time and frequency domains:
- Time domain representation: Speech waveform (time vs amplitude)



- Frequency domain representation: Spectrogram (time vs frequency) – Time-Frequency representation of the speech signal



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Sequence Learning Problem: Spoken Word Recognition

- Representing the speech in time and frequency domains:
  - Time domain representation: Speech waveform (time vs amplitude)
  - Frequency domain representation: Spectrogram (time vs frequency) - Time-Frequency representation of the speech signal
- Features are extracted from frequency domine representation
- The speech signal is a slowly time varying signal
- Quasi-stationary: When examined over a sufficiently short period of time (20-30 ms), its characteristics are fairly stationary
- Over long periods of time the signal characteristics change to reflect the different speech sounds being spoken

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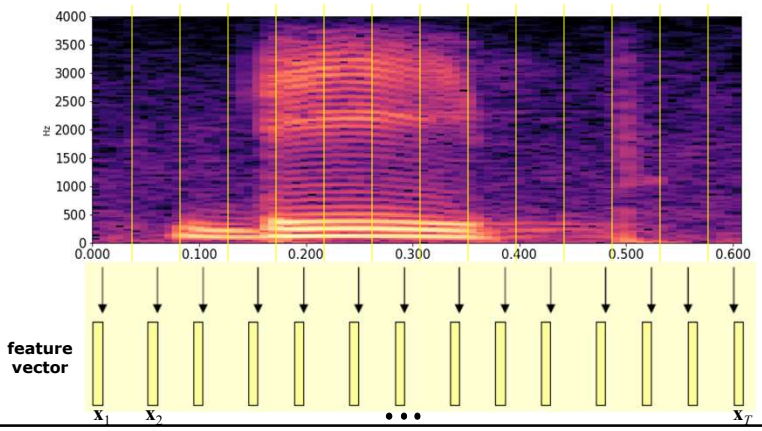
Representation of a Speech Signal for Speech Processing Tasks (Spoken Word Recognition)

- Speech signal represented as a sequence of d-dimensional feature vectors
  - Short time spectral analysis at every window of size 20ms to 30ms with the shift of 10ms to 15ms
  - A d-dimensional feature vector is extracted from each of the window



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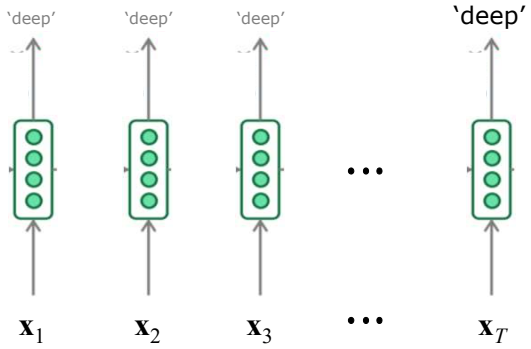
Sequence Learning Problem: Spoken Word Recognition



- The prediction clearly does not depend only on the last feature vector in the sequence but also on some feature vectors which appear before
- As the duration of the utterance varies for each example, the length of the sequence of feature vectors varies
- Network is performing the same task at each step (input: feature vector (speech window), output: spoken word (class))-Don't care about intermediate outputs

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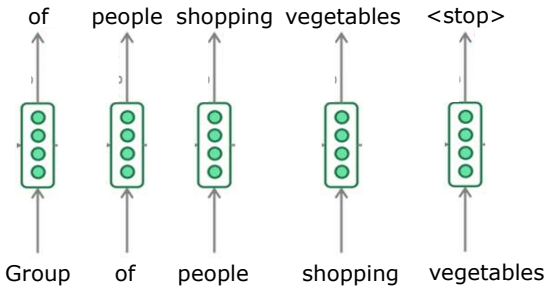
Sequence Learning Problem

- Natural sentence generation
  - Given $t - i$ words predict the t^{th} word
- Each input corresponding to one time step



- Inputs are dependent on each other
- Lengths of the inputs are not fixed (sentences could have arbitrary number of words)
- Each network is performing same task (input: word, output: word)
- **Sequence learning problem**:
  - Look at the sequence of (dependent) inputs and produce output(s)

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Neural Network Model for Sequence Learning Problem

- Model for dealing with sequences should
  - Account for dependence between inputs
  - Account for variable number of inputs
  - Make sure that the function executed at each time step is the same

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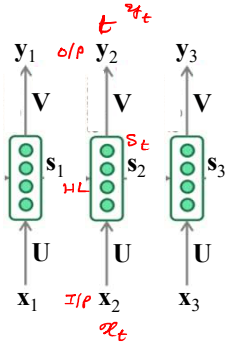
Neural Network Model for Sequence Learning Problem



- Function executed at each time step:

$$\mathbf{s}_t = \Omega(\mathbf{U}\mathbf{x}_t + \mathbf{b})$$
$$\mathbf{y}_t = \Phi(\mathbf{V}\mathbf{s}_t + \mathbf{c})$$

  - t is time step
  - $\mathbf{x}_t \in \mathbb{R}^d$, d is dimension of input variable
  - $\mathbf{y}_t \in \mathbb{R}^K$, K is dimension of output variable (e.g. K classes)
  - $\mathbf{s}_t \in \mathbb{R}^M$, M is number of nodes in hidden layer (last)

- Since the same function to be executed at each time step we should share the same network (i.e., same parameters at each time step)

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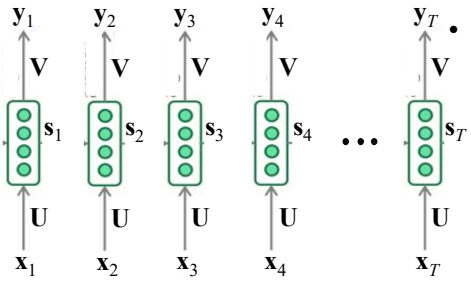
Neural Network Model for Sequence Learning Problem



- Function executed at each time step:

$$\mathbf{s}_t = \Omega(\mathbf{U}\mathbf{x}_t + \mathbf{b})$$
$$\mathbf{y}_t = \Phi(\mathbf{V}\mathbf{s}_t + \mathbf{c})$$

  - t is time step

- Since the same function to be executed at each time step we should share the same network (i.e., same parameters at each time step)
- This parameter sharing also ensures that the network becomes agnostic to the length (size) of the input
- Since same function (with same parameters) is computed at each time step, the number of timesteps doesn't matter
- Just create multiple copies of the network and execute them at each time step

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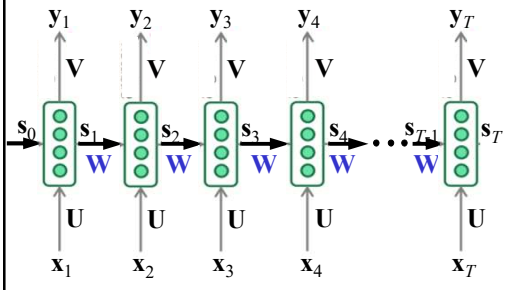
Neural Network Model for Sequence Learning Problem



- How do you account for dependence between inputs?
- Add a recurrent connection in the network to account for dependence between inputs

- Function executed at each time step:

$$\mathbf{s}_t = \Omega(\mathbf{U}\mathbf{x}_t + \mathbf{W}\mathbf{s}_{t-1} + \mathbf{b})$$
$$\mathbf{y}_t = \Phi(\mathbf{V}\mathbf{s}_t + \mathbf{c})$$

- $\mathbf{s}_t$ is the state of the network at time step t
  - $\mathbf{s}_0$ is constant
- The parameters $\mathbf{W}$, $\mathbf{U}$, $\mathbf{V}$, $\mathbf{c}$, $\mathbf{b}$ are shared across time steps

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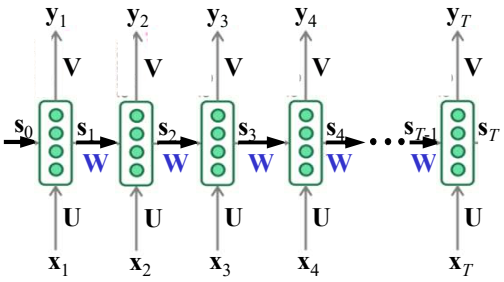
Recurrent Neural Network (RNN)

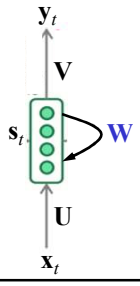


tanh.

$$\mathbf{s}_t = \Omega(\mathbf{U}\mathbf{x}_t + \mathbf{W}\mathbf{s}_{t-1} + \mathbf{b})$$

$$\mathbf{y}_t = \Phi(\mathbf{V}\mathbf{s}_t + \mathbf{c})$$

- More compactly represented as:



$$\mathbf{y}_t = f(\mathbf{x}_t, \mathbf{s}_{t-1}, \mathbf{U}, \mathbf{V}, \mathbf{W}, \mathbf{b}, \mathbf{c})$$

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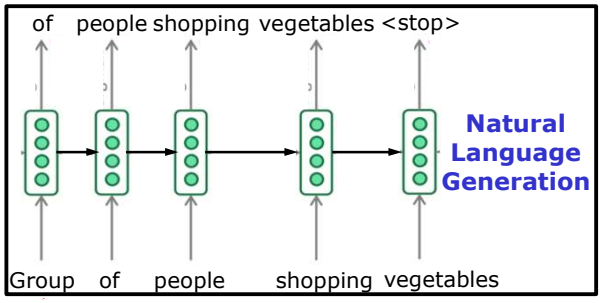
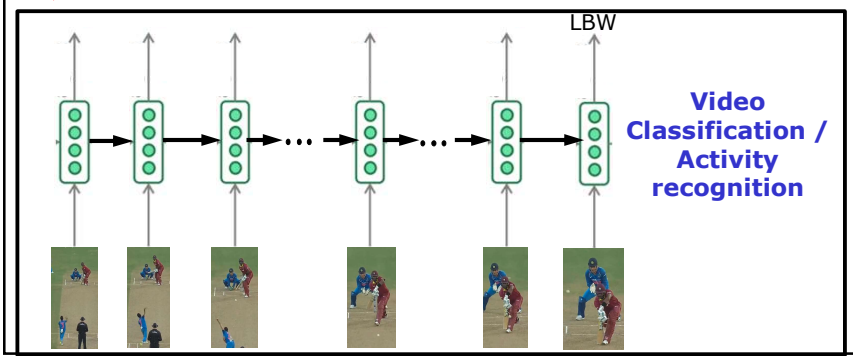
RNN for Sequence Learning Problem



- Recurrent connections between time steps account for dependence between inputs



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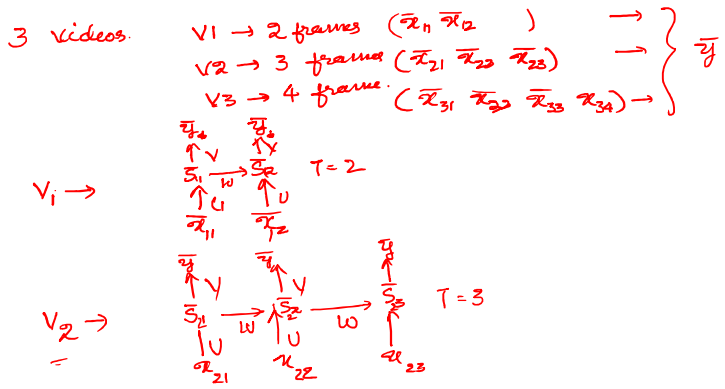
Training RNN



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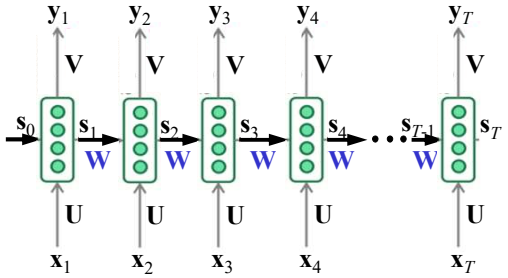
Training RNN: Backpropagation Through Time (BPTT)



- Gradient descent method
- Error backpropagation algorithm
- Forward computation:
  - Innerproduct computation over all the time steps
  - Activation function computation over all the time steps
- Backward operation:
  - Error/Loss calculation and propagation over all previous time steps
  - Modification of weights over all previous time steps

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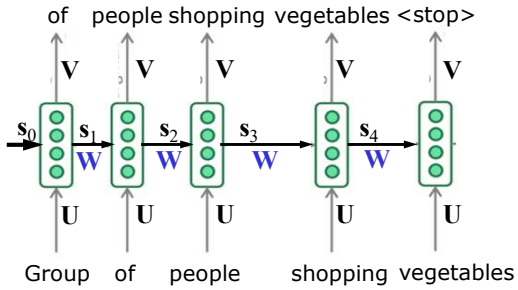
Backpropagation Through Time (BPTT)



- Consider the task of natural language generation (predicting the next word)
- For simplicity assume that there are only 6 words in the vocabulary (Group, of, people, shopping, vegetables, <stop>)
- At each time step we need to predict one of these 6 words
- Suitable output function: Softmax
- Suitable loss function: Cross Entropy

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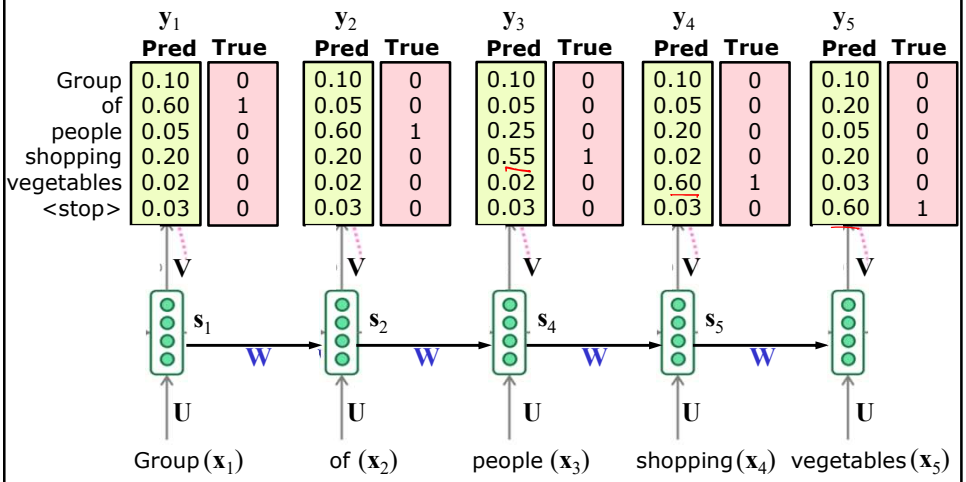
Backpropagation Through Time (BPTT)



- Initialize parameter set $\theta = \{U, V, W, c, b\}$ randomly and the network predicts the probabilities

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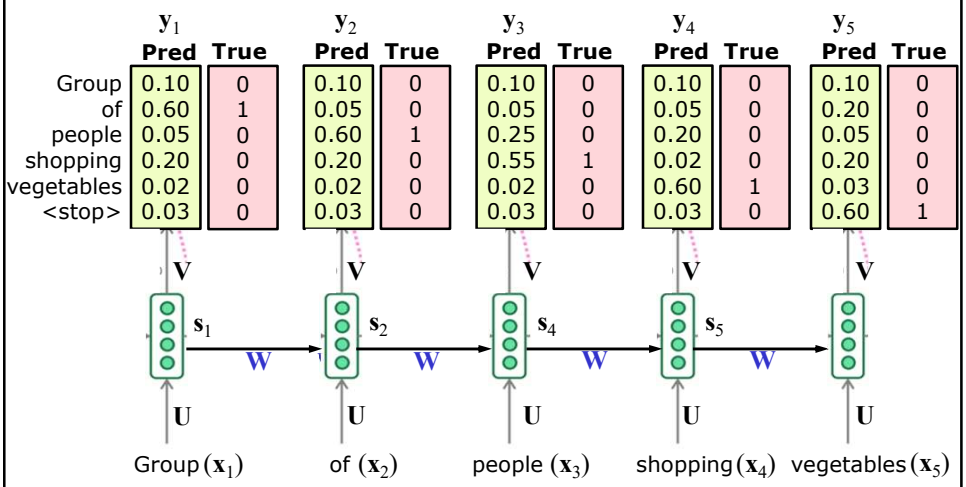
Backpropagation Through Time (BPTT)



- Need to answer following 2 questions:
  - What is the total loss made by the model?
  - How do we backpropagate this loss function and update the parameters of the network

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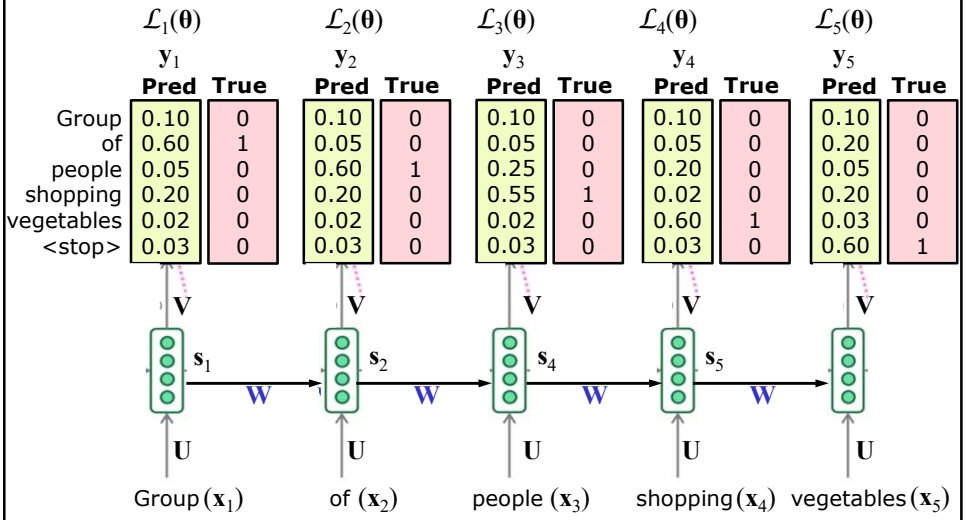
Backpropagation Through Time (BPTT)



- Compute the loss over each time step: $\mathcal{L}_t(\theta)$
- Total loss is the sum of the loss over all time steps

$$\mathcal{L}(\theta) = \sum_{t=1}^{T} \mathcal{L}_t(\theta)$$

T: Number of time steps

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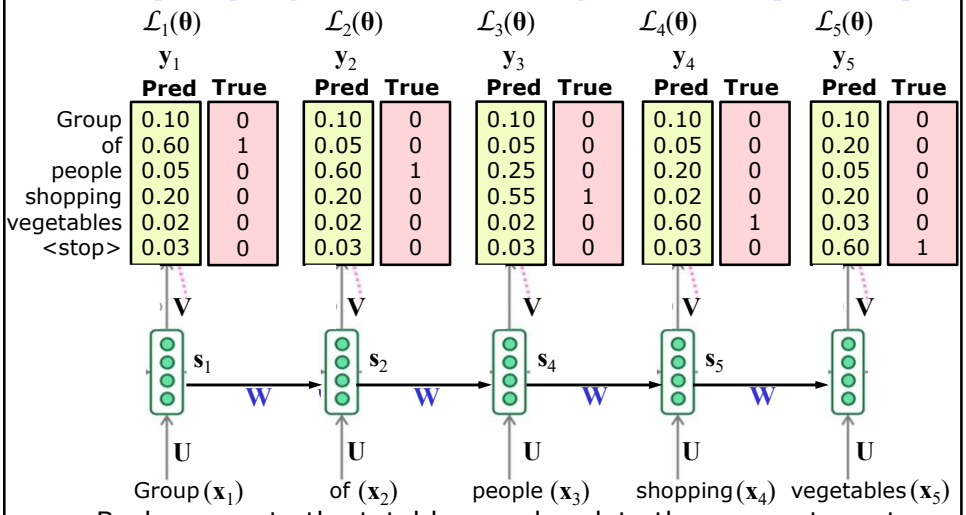
Backpropagation Through Time (BPTT)



- Backpropagate the total loss and update the parameter set θ of the network
- For backpropagation we need to compute the gradients w.r.t. $\mathbf{U}$, $\mathbf{V}$, $\mathbf{W}$, $\mathbf{c}$, $\mathbf{b}$

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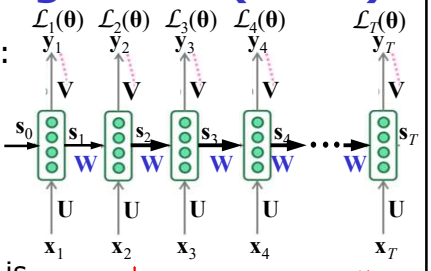
Backpropagation Through Time (BPTT)

- Change in the weight matrix $\mathbf{V}$:

$$\Delta \mathbf{V} = -\eta \nabla_{\mathbf{V}} \mathcal{L}(\boldsymbol{\theta})$$

$$\nabla_{\mathbf{V}} \mathcal{L}(\boldsymbol{\theta}) = \frac{\partial \mathcal{L}(\boldsymbol{\theta})}{\partial \mathbf{V}} = \sum_{t=1}^{T} \frac{\partial \mathcal{L}_t(\boldsymbol{\theta})}{\partial \mathbf{V}}$$



  - Each term in the summation is simply the derivative of the loss w.r.t. the weights in the output layer
  - This gradient computation is same as that of the backpropagation in fully connected neural network (between hidden and output layer)
- Updating the weight $\mathbf{V}$:

$$\mathbf{V} = \mathbf{V} + \Delta \mathbf{V}$$

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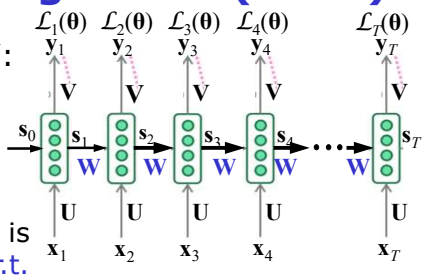
Backpropagation Through Time (BPTT)



- Change in the weight matrix $\mathbf{U}$:

$$\Delta\mathbf{U} = -\eta\nabla_u \mathcal{L}(\boldsymbol{\theta})$$

$$\nabla_u \mathcal{L}(\boldsymbol{\theta}) = \frac{\partial \mathcal{L}(\boldsymbol{\theta})}{\partial \mathbf{U}} = \sum_{t=1}^{T} \frac{\partial \mathcal{L}_t(\boldsymbol{\theta})}{\partial \mathbf{U}}$$

  - Each term is the summation is the derivative of the loss w.r.t. the weights in the input/hidden layer
  - This gradient computation is same as that of the backpropagation in fully connected neural network (between hidden layers and hidden layer to input layer)
  - By chain rule, assuming single hidden layer

$$\frac{\partial \mathcal{L}_t(\boldsymbol{\theta})}{\partial \mathbf{U}} = \frac{\partial \mathcal{L}_t(\boldsymbol{\theta})}{\partial \mathbf{y}_t}\frac{\partial \mathbf{y}_t}{\partial \mathbf{a}_{2t}}\frac{\partial \mathbf{a}_{2t}}{\partial \mathbf{s}_t}\frac{\partial \mathbf{s}_t}{\partial \mathbf{a}_{1t}}\frac{\partial \mathbf{a}_{1t}}{\partial \mathbf{U}} \qquad \mathbf{y}_t = \Phi(\mathbf{a}_{2t}) \quad \mathbf{a}_{2t} = \mathbf{V}\mathbf{s}_t + \mathbf{c}$$

$$\mathbf{s}_t = \Omega(\mathbf{a}_{1t}) \quad \mathbf{a}_{1t} = \mathbf{U}\mathbf{x}_t + \mathbf{W}\mathbf{s}_{t-1} + \mathbf{b}$$

- Updating the weight $\mathbf{U}$: $\mathbf{U} = \mathbf{U} + \Delta\mathbf{U}$

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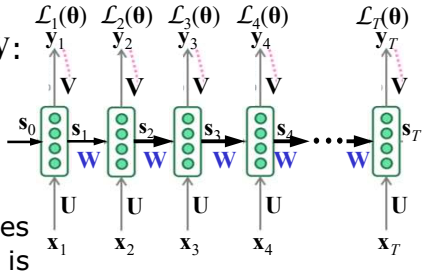
Backpropagation Through Time (BPTT)



- Change in the weight matrix $\mathbf{W}$:

$$\Delta\mathbf{W} = -\eta\nabla_w \mathcal{L}(\boldsymbol{\theta})$$

$$\nabla_w \mathcal{L}(\boldsymbol{\theta}) = \frac{\partial \mathcal{L}(\boldsymbol{\theta})}{\partial \mathbf{W}} = \sum_{t=1}^{T} \frac{\partial \mathcal{L}_t(\boldsymbol{\theta})}{\partial \mathbf{W}}$$

  - By the chain rule of derivatives this gradient at time sept t is obtained by summing gradients along all the paths from $\mathcal{L}_t(\boldsymbol{\theta})$ to $\mathbf{W}$
  - Paths connecting from $\mathcal{L}_t(\boldsymbol{\theta})$ to $\mathbf{W}$:
    - $\mathcal{L}_t(\boldsymbol{\theta})$ depends on $\mathbf{s}_t$
    - $\mathbf{s}_t$ in turn depends on $\mathbf{s}_{t-1}$ and $\mathbf{W}$
    - $\mathbf{s}_{t-1}$ in turn depends on $\mathbf{s}_{t-2}$ and $\mathbf{W}$
    - ...
    - $\mathbf{s}_2$ in turn depends on $\mathbf{s}_1$ and $\mathbf{W}$
    - $\mathbf{s}_1$ in turn depends on $\mathbf{s}_0$ and $\mathbf{W}$ ($\mathbf{s}_0$ is constant starting state)

Ordered network

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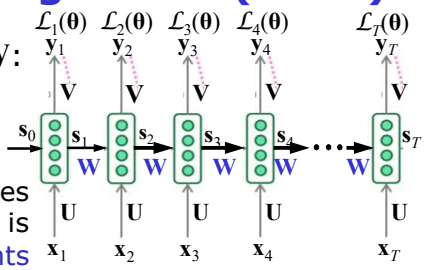
Backpropagation Through Time (BPTT)

- Change in the weight matrix $\mathbf{W}$:

$$\nabla_w \mathcal{L}(\boldsymbol{\theta}) = \frac{\partial \mathcal{L}(\boldsymbol{\theta})}{\partial \mathbf{W}} = \sum_{t=1}^{T} \frac{\partial \mathcal{L}_t(\boldsymbol{\theta})}{\partial \mathbf{W}}$$

  - By the chain rule of derivatives this gradient at time sept t is obtained by summing gradients along all the paths from $\mathcal{L}_t(\boldsymbol{\theta})$ to $\mathbf{W}$

$$\frac{\partial \mathcal{L}_t(\boldsymbol{\theta})}{\partial \mathbf{W}} = \frac{\partial \mathcal{L}_t(\boldsymbol{\theta})}{\partial \mathbf{s}_t} \frac{\partial \mathbf{s}_t}{\partial \mathbf{W}}$$



$$\mathbf{s}_t = \Omega(\mathbf{U}\mathbf{x}_t + \mathbf{W}\mathbf{s}_{t-1} + \mathbf{b})$$

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Backpropagation Through Time (BPTT)

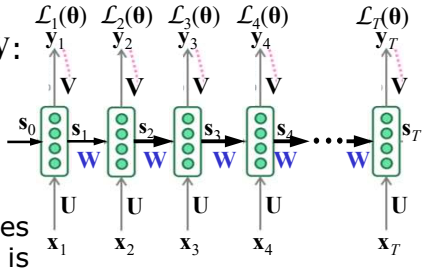
- Change in the weight matrix $\mathbf{W}$:

$$\Delta \mathbf{W} = -\eta \nabla_w \mathcal{L}(\boldsymbol{\theta})$$

$$\nabla_w \mathcal{L}(\boldsymbol{\theta}) = \frac{\partial \mathcal{L}(\boldsymbol{\theta})}{\partial \mathbf{W}} = \sum_{t=1}^{T} \frac{\partial \mathcal{L}_t(\boldsymbol{\theta})}{\partial \mathbf{W}}$$

  - By the chain rule of derivatives this gradient at time sept t is obtained by summing gradients along all the paths from $\mathcal{L}_t(\boldsymbol{\theta})$ to $\mathbf{W}$

$$\frac{\partial \mathcal{L}_t(\boldsymbol{\theta})}{\partial \mathbf{W}} = \frac{\partial \mathcal{L}_t(\boldsymbol{\theta})}{\partial \mathbf{s}_t} \left(\sum_{k=1}^{t} \frac{\partial \mathbf{s}_t}{\partial \mathbf{s}_k} \frac{\partial^+ \mathbf{s}_k}{\partial \mathbf{W}} \right)$$



$$\mathbf{s}_t = \Omega(\mathbf{U}\mathbf{x}_t + \mathbf{W}\mathbf{s}_{t-1} + \mathbf{b})$$

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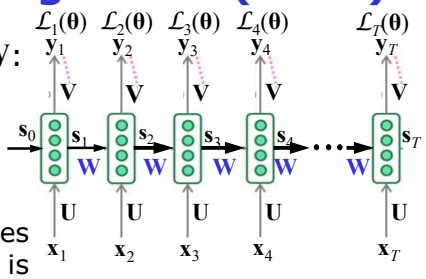
Backpropagation Through Time (BPTT)



- Change in the weight matrix $\mathbf{W}$:

$$\Delta\mathbf{W} = -\eta\nabla_w\mathcal{L}(\boldsymbol{\theta})$$

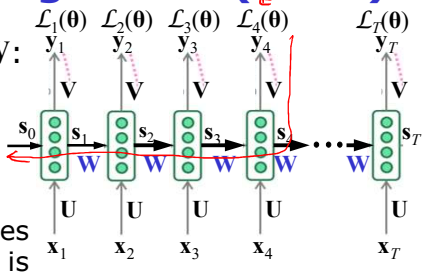
$$\nabla_w\mathcal{L}(\boldsymbol{\theta}) = \frac{\partial\mathcal{L}(\boldsymbol{\theta})}{\partial\mathbf{W}} = \sum_{t=1}^{T}\frac{\partial\mathcal{L}_t(\boldsymbol{\theta})}{\partial\mathbf{W}}$$

  - By the chain rule of derivatives this gradient at time sept t is obtained by summing gradients along all the paths from $\mathcal{L}_t(\boldsymbol{\theta})$ to $\mathbf{W}$

$$\nabla_w\mathcal{L}(\boldsymbol{\theta}) = \frac{\partial\mathcal{L}(\boldsymbol{\theta})}{\partial\mathbf{W}} = \sum_{t=1}^{T}\frac{\partial\mathcal{L}_t(\boldsymbol{\theta})}{\partial\mathbf{s}_t}\left(\sum_{k=1}^{t}\frac{\partial\mathbf{s}_t}{\partial\mathbf{s}_k}\frac{\partial^+\mathbf{s}_k}{\partial\mathbf{W}}\right)$$

$\frac{\partial^+\mathbf{s}_k}{\partial\mathbf{W}}$ is a tensor

Backpropagation Through Time (BPTT)



- Change in the weight matrix $\mathbf{W}$:

$$\Delta\mathbf{W} = -\eta\nabla_w\mathcal{L}(\boldsymbol{\theta})$$

$$\nabla_w\mathcal{L}(\boldsymbol{\theta}) = \frac{\partial\mathcal{L}(\boldsymbol{\theta})}{\partial\mathbf{W}} = \sum_{t=1}^{T}\frac{\partial\mathcal{L}_t(\boldsymbol{\theta})}{\partial\mathbf{W}}$$

  - By the chain rule of derivatives this gradient at time sept t is obtained by summing gradients along all the paths from $\mathcal{L}_t(\boldsymbol{\theta})$ to $\mathbf{W}$

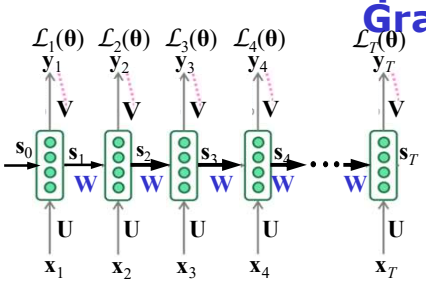
$$\nabla_w\mathcal{L}(\boldsymbol{\theta}) = \frac{\partial\mathcal{L}(\boldsymbol{\theta})}{\partial\mathbf{W}} = \sum_{t=1}^{T}\frac{\partial\mathcal{L}_t(\boldsymbol{\theta})}{\partial\mathbf{s}_t}\left(\sum_{k=1}^{t}\frac{\partial\mathbf{s}_t}{\partial\mathbf{s}_k}\frac{\partial^+\mathbf{s}_k}{\partial\mathbf{W}}\right)$$

$\frac{\partial^+\mathbf{s}_k}{\partial\mathbf{W}}$ is a tensor

- Updating the weight $\mathbf{W}$: Backpropagate over all previous time steps

$$\mathbf{W} = \mathbf{W} + \Delta\mathbf{W}$$

Problem of Exploding and Vanishing Gradients



- Consider the gradient with respect to recurrent connection weights

$$\nabla_w \mathcal{L}(\theta) = \frac{\partial \mathcal{L}(\theta)}{\partial \mathbf{W}} = \sum_{t=1}^{T} \frac{\partial \mathcal{L}_t(\theta)}{\partial \mathbf{W}}$$

$$\frac{\partial \mathcal{L}_t(\theta)}{\partial \mathbf{W}} = \frac{\partial \mathcal{L}_t(\theta)}{\partial \mathbf{s}_t}\left(\sum_{k=1}^{t} \frac{\partial \mathbf{s}_t}{\partial \mathbf{s}_k}\frac{\partial^+ \mathbf{s}_k}{\partial \mathbf{W}}\right)$$

$$\mathbf{s}_j = \Omega(\mathbf{a}_j) \qquad \mathbf{a}_j = \mathbf{U}\mathbf{x}_j + \mathbf{W}\mathbf{s}_{j-1} + b$$

$$\frac{\partial \mathbf{s}_j}{\partial \mathbf{s}_{j-1}} = \frac{\partial \mathbf{s}_j}{\partial \mathbf{a}_j}\frac{\partial \mathbf{a}_j}{\partial \mathbf{s}_{j-1}}$$

$$\frac{\partial \mathbf{s}_j}{\partial \mathbf{s}_{j-1}} = \frac{\partial \Omega(\mathbf{a}_j)}{\partial \mathbf{a}_j}\frac{\partial \mathbf{a}_j}{\partial \mathbf{s}_{j-1}}$$

$$\frac{\partial \mathbf{s}_t}{\partial \mathbf{s}_k} = \frac{\partial \mathbf{s}_t}{\partial \mathbf{s}_{t-1}}\frac{\partial \mathbf{s}_{t-1}}{\partial \mathbf{s}_{t-2}}\cdots\frac{\partial \mathbf{s}_{k+1}}{\partial \mathbf{s}_k} = \prod_{j=k+1}^{t}\frac{\partial \mathbf{s}_j}{\partial \mathbf{s}_{j-1}}$$

$$\frac{\partial \mathbf{s}_t}{\partial \mathbf{s}_k} = \prod_{j=k+1}^{t}\frac{\partial \Omega(\mathbf{a}_j)}{\partial \mathbf{a}_j}\frac{\partial \mathbf{a}_j}{\partial \mathbf{s}_{j-1}}$$

$$\frac{\partial \mathcal{L}_t(\theta)}{\partial \mathbf{W}} = \frac{\partial \mathcal{L}_t(\theta)}{\partial \mathbf{s}_t}\left(\sum_{k=1}^{t}\left[\prod_{j=k+1}^{t}\frac{\partial \Omega(\mathbf{a}_j)}{\partial \mathbf{a}_j}\frac{\partial \mathbf{a}_j}{\partial \mathbf{s}_{j-1}}\right]\frac{\partial^+ \mathbf{s}_k}{\partial \mathbf{W}}\right)$$

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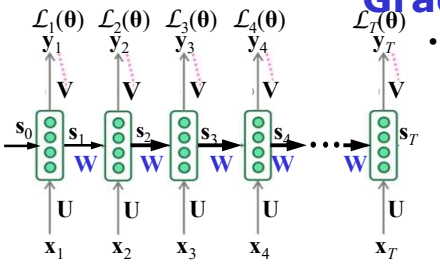
Problem of Exploding and Vanishing Gradients



- Consider the gradient with respect to recurrent connection weights

$$\nabla_w \mathcal{L}(\theta) = \frac{\partial \mathcal{L}(\theta)}{\partial \mathbf{W}} = \sum_{t=1}^{T} \frac{\partial \mathcal{L}_t(\theta)}{\partial \mathbf{W}}$$

$$\frac{\partial \mathcal{L}_t(\theta)}{\partial \mathbf{W}} = \frac{\partial \mathcal{L}_t(\theta)}{\partial \mathbf{s}_t}\left(\sum_{k=1}^{t} \frac{\partial \mathbf{s}_t}{\partial \mathbf{s}_k}\frac{\partial^+ \mathbf{s}_k}{\partial \mathbf{W}}\right)$$

$$\frac{\partial \mathcal{L}_t(\theta)}{\partial \mathbf{W}} = \frac{\partial \mathcal{L}_t(\theta)}{\partial \mathbf{s}_t}\left(\sum_{k=1}^{t}\left[\prod_{j=k+1}^{t}\frac{\partial \Omega(\mathbf{a}_j)}{\partial \mathbf{a}_j}\frac{\partial \mathbf{a}_j}{\partial \mathbf{s}_{j-1}}\right]\frac{\partial^+ \mathbf{s}_k}{\partial \mathbf{W}}\right)$$

If $\Omega(a) \Rightarrow$ logistic $\rightarrow$ range is bounded b/t 0 & 1

$\Omega'(a) = \Omega(a)(1-\Omega(a)) \rightarrow$ range bounded b/t 0 to 0.25

$a = \pm\infty \rightarrow \Omega'(a) = 0$

$a = 0 \rightarrow \Omega'(a) = (0.5)(1-0.5) = 0.25$

$\Omega(a) \rightarrow$ tanh $\rightarrow$ -1 to $+1$

$\Omega'(a) \rightarrow 1 - \Omega^2(a) \rightarrow 0$ to 1

$t \approx T \quad k = 1$

$\frac{\partial \mathbf{s}_t}{\partial \mathbf{s}_k} \rightarrow 0$

$\frac{\partial \mathcal{L}_t(\theta)}{\partial \mathbf{W}} \rightarrow 0$

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Problem of Exploding and Vanishing Gradients

- One simple way of avoiding this is to use truncated backpropogation where we restrict the product to less than $t-k$ terms
- Other ways to avoid exploding and vanishing gradient is by using LSTM and GRU

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Summery: RNN

- RNN used for sequential learning problem
  - Each input is dependent on the previous or future input
  - In many applications the input is not of a fixed size
- RNN uses backpropagation through time for training
- RNN may suffer from vanishing/exploding gradients problem
- This issue is addressed in long short term memory (LSTM) model
  - LSTM is a special case of RNN

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Text Books

1. Ian Goodfellow, Yoshua Bengio and Aaron Courville, Deep learning, MIT Press, Available online: http://www.deeplearningbook.org, 2016
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4. Satish Kumar, Neural Networks - A Class Room Approach, Second Edition, Tata McGraw-Hill, 2013.
5. S. Haykin, Neural Networks and Learning Machines, Prentice Hall of India, 2010.
6. C. M. Bishop, *Pattern Recognition and Machine Learning*, Springer, 2006.
7. J. Han and M. Kamber, *Data Mining: Concepts and Techniques*, Third Edition, Morgan Kaufmann Publishers, 2011.
8. S. Theodoridis and K. Koutroumbas, *Pattern Recognition*, Academic Press, 2009.

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